

Objectives

- Compare and contrast different ways, O₂ and CO₂ are transported in blood. How does Henry's Law apply to those?
- Compare and contrast structure, function and pathologic implications of adult and fetal hemoglobin, methemoglobin and sickle cell hemoglobin.
- Explain the differences between oxygen content and binding capacity.
- Explain how cooperative oxygen binding influences the sigmoidal shape of the oxygen dissociation curve. How is this shape beneficial to oxidative metabolism?
- Explain the physiological rationale behind PCO₂, pH, temperature, and 2,3-DPG being able to shift the oxygen dissociation curve (left or right). What conditions favor the 'Bohr shift' and why?
- Explain how pulse oximetry works. How does it focus on arterial and not venous or capillary blood?
- Explain the effect of CO on hemoglobin. What are the consequences and how do they matter?
- Compare and contrast the three ways CO₂ can be transported in blood. How does the Haldane effect facilitate normal metabolism?

HBS: Sickle Cell Hemoglobin

Lower O₂ Affinity

Normal hemoglobin β chain

Valine — Histidine — Leucine — Threonine — Proline — Glutamic acid — Glutamic acid

Valine — Histidine — Leucine — Threonine — Proline — Valine — Glutamic acid

Sickle cell anemia hemoglobin β chain



O₂ Binding Capacity vs. O₂ Content

The O₂ -binding capacity is the **maximum amount of O₂** that can be **bound to hemoglobin** per volume of blood

1 g of hemoglobin A can bind 1.34 mL O₂

The normal concentration of hemoglobin A in blood is 15 g/100 mL

20.1 mL O₂ /100 mL blood maximally bound to hemoglobin
(15 g/100 mL × 1.34 mL O₂ /g hemoglobin)

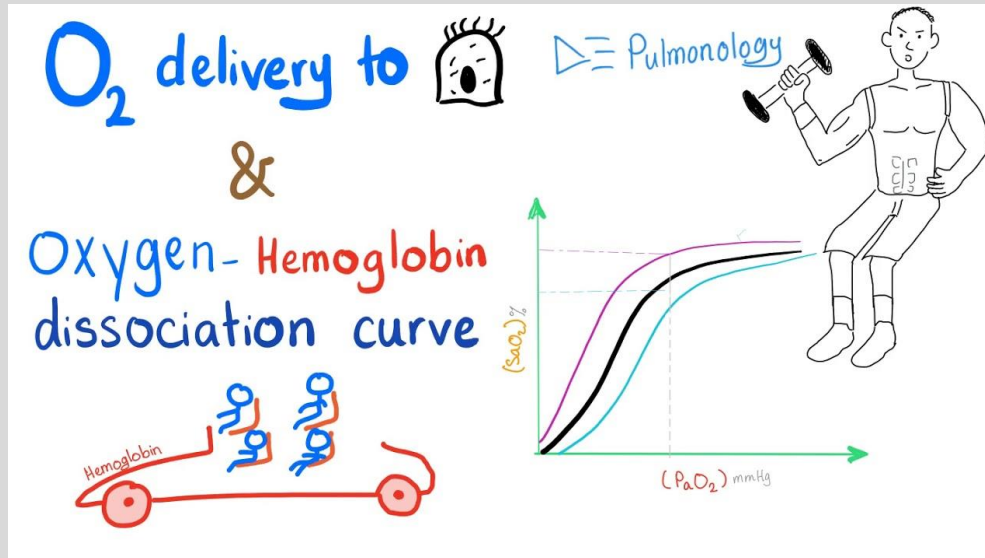
The O₂ content is the **actual amount of O₂ per volume of blood**

$$\text{O}_2 \text{ content} = (\text{O}_2\text{-binding capacity} \times \% \text{Saturation}) + \text{Dissolved O}_2$$

O₂ Delivery to Tissues

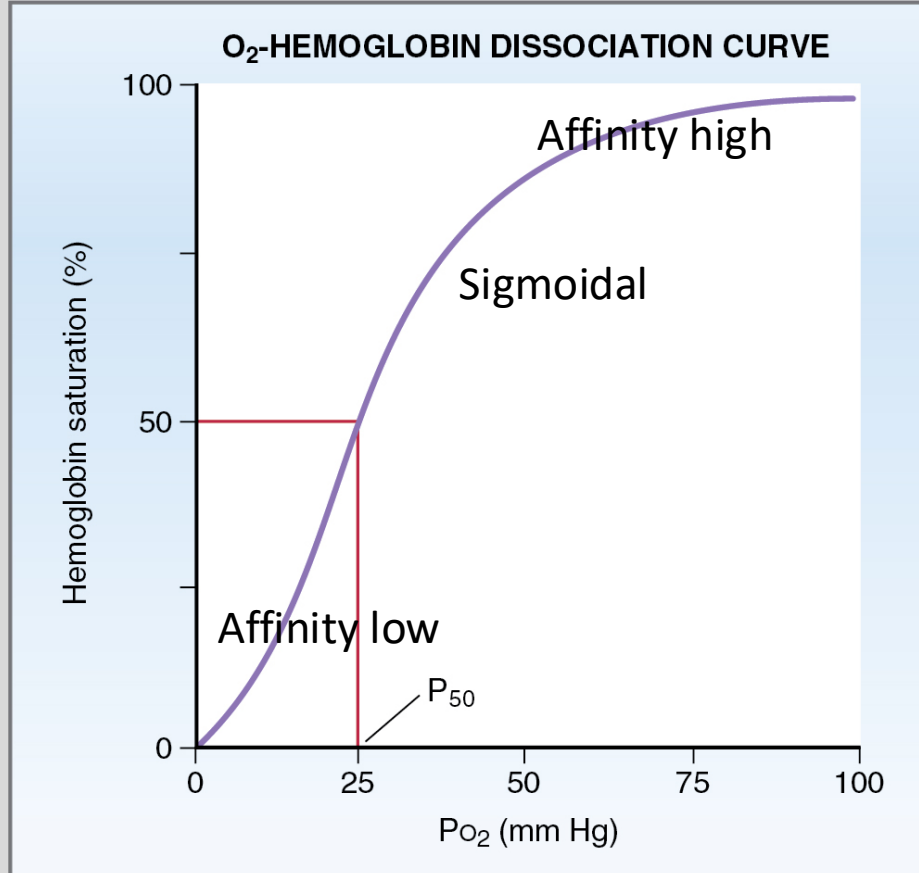
O₂ content of blood is the sum of **dissolved O₂ (~2%)** and **O₂-hemoglobin (~98%)**

$$\begin{aligned}\text{O}_2 \text{ delivery} &= \text{Cardiac output} \times \text{O}_2 \text{ content of blood} \\ &= \text{Cardiac output} \times (\text{Dissolved O}_2 + \text{O}_2\text{-hemoglobin})\end{aligned}$$



O₂ – Hemoglobin Dissociation Curve

Changes in Affinity through positive Cooperativity

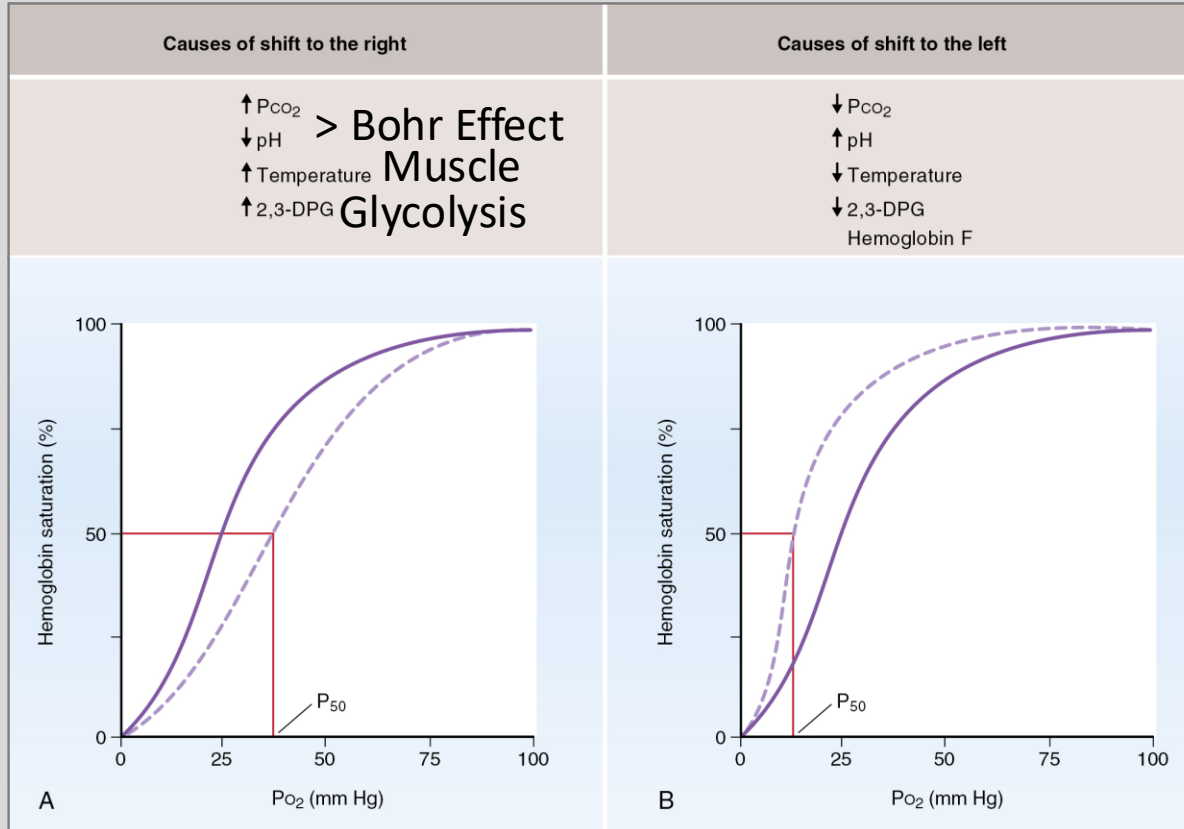


P o ₂ (mm Hg)	Saturation (%)
10	25
20	35
25	50
30	60
40	75
50	85
60	90
80	96
100	98

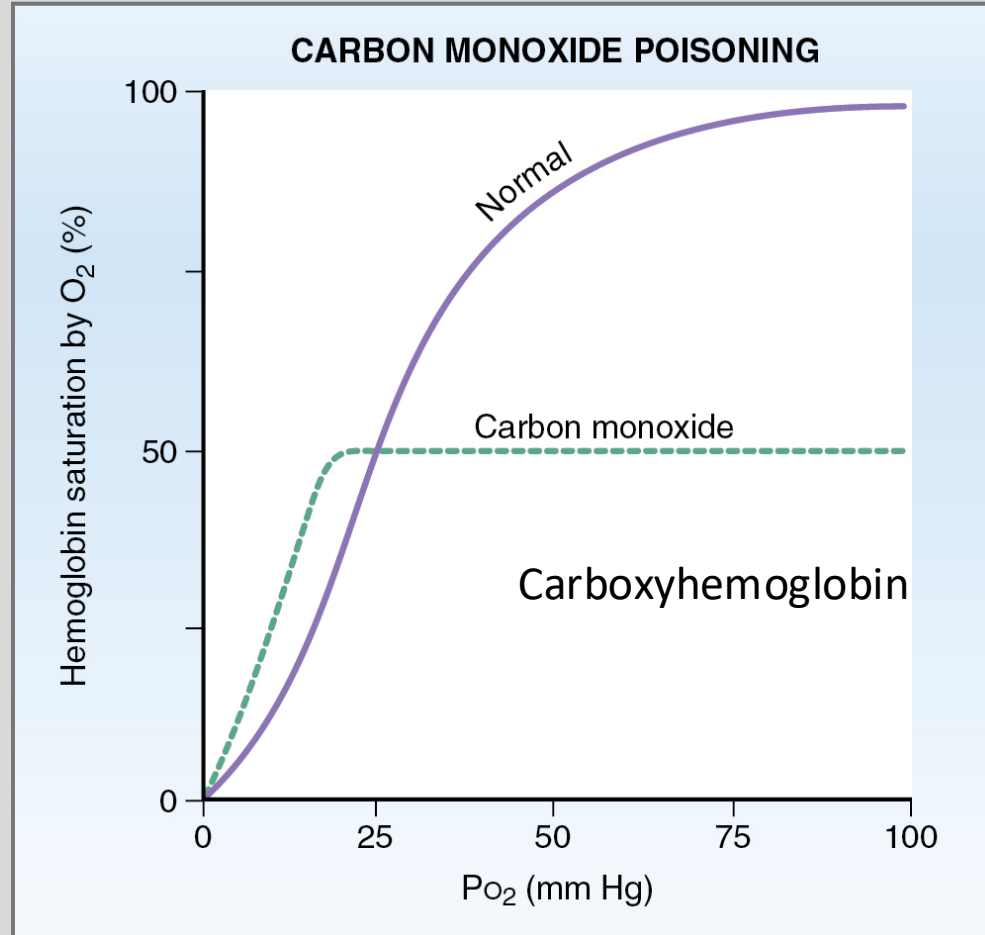
The P o₂ that corresponds to 50% saturation of hemoglobin is called P₅₀.

Changes in the O₂-Hemoglobin Dissociation Curve

Metabolism & its Products



The Effect of CO – Carbon Monoxide

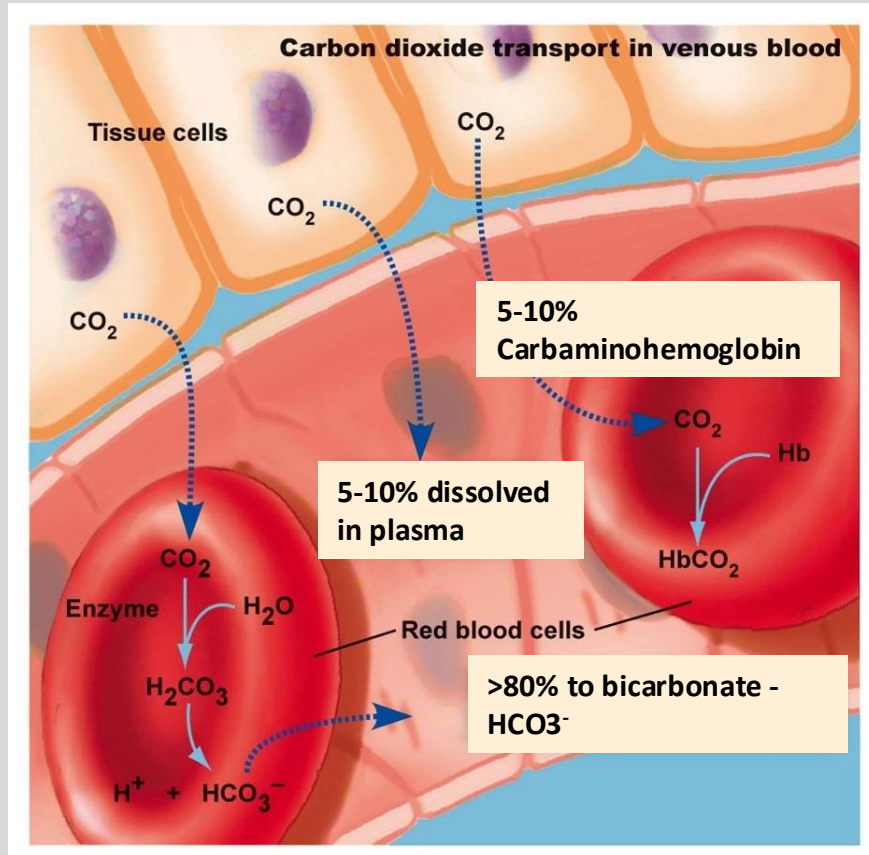


- 210-250 times higher affinity to heme than O₂

Therapies:

- Regular air: ~5h
- Air saturated with O₂: ~1.5h
- Hyperbaric chamber (3ATM) ~20min

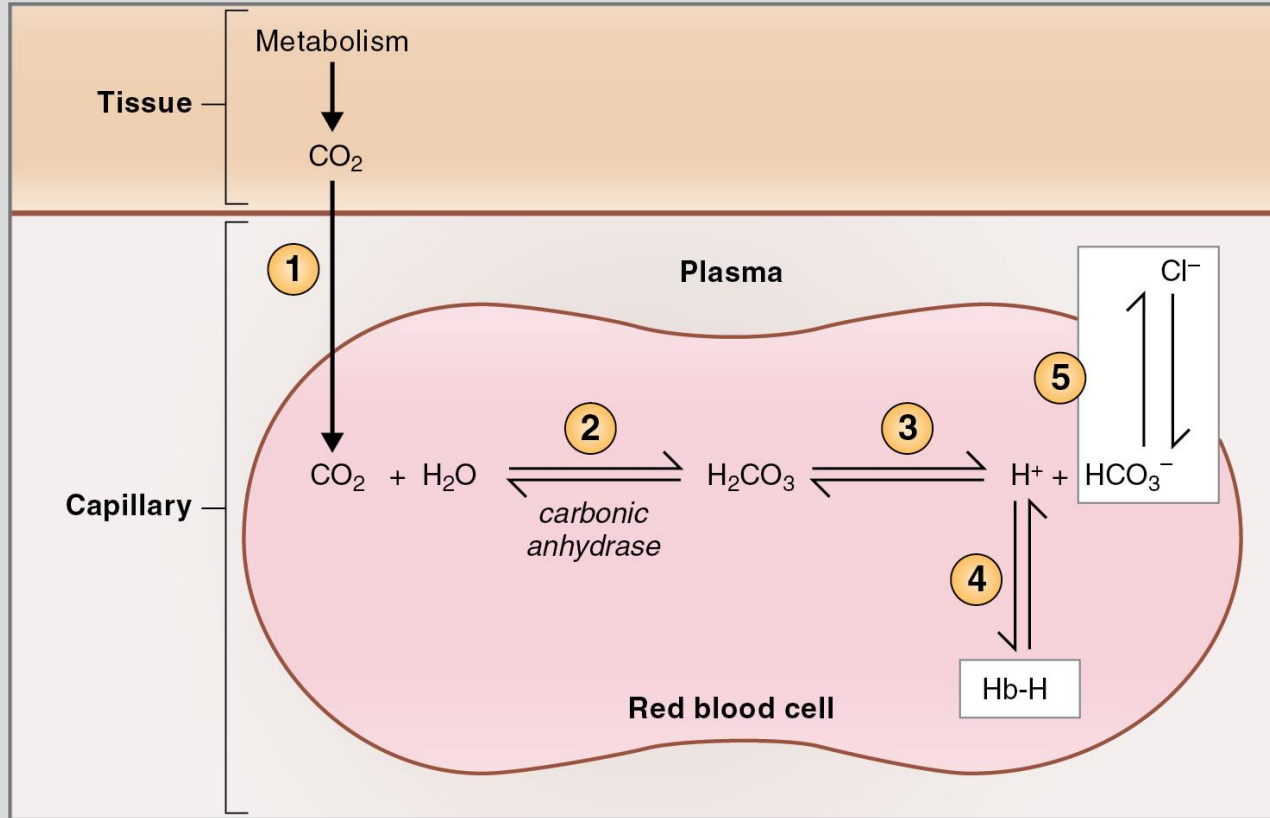
CO₂ Dissolved in Blood (Free)



Percentages differ
somewhat between texts.
Stick with the values here!

2.8 mL CO₂ /100 mL blood
(40 mm Hg × 0.07 mL CO₂
/100 mL blood per mm Hg)
→ that is ~5%

Bicarbonate HCO_3^-



- **Chloride shift** via band-3 protein used for anion exchange
- Deoxy-Hgb: better buffer for H^+

The reverse reactions happen in lung tissue where CO_2 and H_2O are released!

Medical Board-Style Practice Question

A 19-year-old college track athlete is competing in a high-intensity 400-meter sprint. Midway through the race, his actively contracting skeletal muscle cells experience a sharp increase in metabolic rate, rapidly producing large quantities of carbon dioxide (CO_2) and lactic acid, while generating local heat (hyperthermia). Which of the following best describes the physiological shift occurring in the hemoglobin-oxygen dissociation curve within this localized muscle tissue, and what is its functional benefit?

- A. A leftward shift, which increases hemoglobin's affinity for oxygen to ensure the muscle tissue absorbs gas faster.
- B. A rightward shift, which decreases hemoglobin's affinity for oxygen, facilitating its unloading to the oxygen-starved muscle.
- C. A leftward shift, which stabilizes the relaxed (R) state of hemoglobin to prevent the premature release of oxygen.
- D. A rightward shift, which binds oxygen more tightly to hemoglobin to protect the red blood cells from acidic denaturation.
- E. No shift occurs, because systemic baroreceptors automatically adjust cardiac output to keep local tissue curves completely constant.